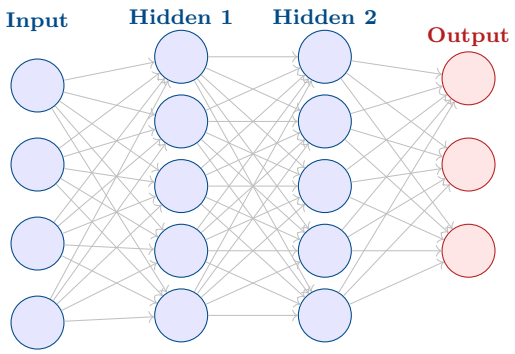
Advanced Machine Learning

MSc 2-Credit Course — Complete Study Notes



Topics Covered:

Supervised & Unsupervised Learning · Multi-Task Learning · Bayesian Modelling · Gaussian Processes · Randomized Methods · Bayesian Neural Networks · Approximate Inference · Variational Autoencoders · Generative Models · RNN / BPTT / LSTM · Attention & Memory Networks · Neural Turing Machines · Machine Translation · Question Answering · Speech Recognition · Syntactic & Semantic Parsing · GPU Optimisation

Department of Computer Science
MSc in Machine Learning & AI

March 31, 2026

Contents

Part I

Core Learning Paradigms

Chapter 1

Supervised, Unsupervised, and Multi-Task Learning

1.1 Supervised Learning

Definition 1.1.1: Supervised Learning

Given a labelled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, supervised learning finds a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ that minimises a loss $\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^N \ell(f_\theta(\mathbf{x}_i), y_i)$.

1.1.1 Empirical Risk Minimisation (ERM)

The goal is to minimise the expected risk:

$$R(f) = \mathbb{E}_{(\mathbf{x},y)\sim p_{\text{data}}}[\ell(f(\mathbf{x}), y)] \approx \hat{R}(f) = \frac{1}{N}\sum_{i=1}^{N} \ell(f(\mathbf{x}_i), y_i).$$

Regularisation (L1/L2) controls overfitting:

$$\mathcal{L}_{\text{reg}}(\theta) = \hat{R}(f) + \lambda\|\theta\|_p^p, \quad p \in \{1, 2\}.$$

1.1.2 Key Algorithms

- **Linear & Logistic Regression** — closed-form or gradient descent.
- **Support Vector Machines** — maximise margin; kernel trick for non-linearity.
- **Decision Trees / Random Forests / Gradient Boosting** — ensemble methods.
- **Deep Neural Networks** — universal function approximators trained via SGD.

1.2 Unsupervised Learning

Definition 1.2.1: Unsupervised Learning

Learning structure from unlabelled data $\{\mathbf{x}_i\}_{i=1}^N$ by modelling the data distribution $p(\mathbf{x})$ or finding compact representations.

1.2.1 Clustering

K-Means minimises the within-cluster sum of squares:

$$\min_{\{\mu_k\}} \sum_{k=1}^{K} \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \mu_k\|^2.$$

Gaussian Mixture Models (GMM) provide a probabilistic generalisation:

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k), \quad \sum_k \pi_k = 1.$$

Parameters are estimated via the *Expectation-Maximisation (EM)* algorithm.

1.2.2 Dimensionality Reduction

Principal Component Analysis (PCA) projects data onto the top-d eigenvectors of the covariance matrix $\Sigma = \frac{1}{N}\mathbf{X}^\top \mathbf{X}$. **Autoencoders** learn a low-dimensional code $\mathbf{z} = \text{enc}_\phi(\mathbf{x})$ and reconstruction $\hat{\mathbf{x}} = \text{dec}_\theta(\mathbf{z})$.

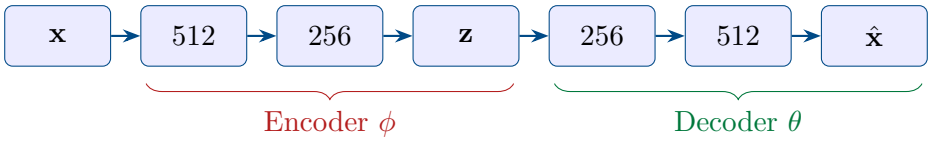


Figure 1.1: Standard Autoencoder architecture: encoder compresses $\mathbf{x}$ to latent code $\mathbf{z}$; decoder reconstructs $\hat{\mathbf{x}}$.

1.3 Multi-Task Learning

Definition 1.3.1: Multi-Task Learning (MTL)

MTL trains a model jointly on T related tasks, sharing representations to improve generalisation. The objective is:

$$\min_{\theta} \sum_{t=1}^{T} w_t \, \mathcal{L}_t(\theta_{\text{shared}}, \theta_t)$$

where θ_{shared} is a common backbone and θ_t are task-specific heads.

1.3.1 Hard vs. Soft Parameter Sharing

- **Hard sharing:** All tasks share hidden layers; only output layers differ.
- **Soft sharing:** Each task has its own network; parameters are regularised to stay close (e.g. $\|\theta_t - \theta_{t'}\|^2 \leq \epsilon$).
- **Cross-stitch networks:** Learn linear combinations of activations across tasks at each layer.

Part II

Bayesian and Probabilistic Methods

Chapter 2

Bayesian Modelling and Gaussian Processes

2.1 Bayesian Inference Fundamentals

Definition 2.1.1: Bayes' Theorem

$$\underbrace{p(\theta \mid \mathcal{D})}_{\text{posterior}} = \frac{\overbrace{p(\mathcal{D} \mid \theta)}^{\text{likelihood}}\ \overbrace{p(\theta)}^{\text{prior}}}{\underbrace{p(\mathcal{D})}_{\text{marginal likelihood}}}$$

The posterior captures our updated belief about parameters θ after observing data $\mathcal{D}$.

2.1.1 Predictive Distribution

Rather than a point estimate, Bayesian methods give:

$$p(y^* \mid \mathbf{x}^*, \mathcal{D}) = \int p(y^* \mid \mathbf{x}^*, \theta)\, p(\theta \mid \mathcal{D})\, d\theta.$$

This integral is generally intractable and motivates approximate inference.

2.2 Gaussian Processes

Definition 2.2.1: Gaussian Process

A Gaussian Process (GP) is a collection of random variables, any finite number of which are jointly Gaussian. It is fully specified by a mean function $m(\mathbf{x})$ and a covariance (kernel) function $k(\mathbf{x}, \mathbf{x}')$:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}),\ k(\mathbf{x}, \mathbf{x}'))\,.$$

2.2.1 GP Regression

Given observations $\mathbf{y} = f(\mathbf{X}) + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$, the posterior predictive at test points $\mathbf{X}_*$ is:

$$\mu_* = \mathbf{K}_{*\mathbf{X}} \left(\mathbf{K}_{\mathbf{XX}} + \sigma_n^2 \mathbf{I} \right)^{-1} \mathbf{y}, \tag{2.1}$$

$$\Sigma_* = \mathbf{K}_{**} - \mathbf{K}_{*\mathbf{X}} \left(\mathbf{K}_{\mathbf{XX}} + \sigma_n^2 \mathbf{I} \right)^{-1} \mathbf{K}_{\mathbf{X}*}, \tag{2.2}$$

where $[\mathbf{K}_{\mathbf{XX}}]_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$.

2.2.2 Kernel Functions

Kernel	Formula	Property
Squared Exponential (RBF)	$\exp\left(-\frac{\|\mathbf{x}-\mathbf{x}'\|^2}{2\ell^2}\right)$	Infinitely differentiable
Matérn 5/2	$\left(1 + \frac{\sqrt{5}r}{\ell} + \frac{5r^2}{3\ell^2}\right) e^{-\frac{\sqrt{5}r}{\ell}}$	Twice differentiable
Periodic	$\exp\left(-\frac{2\sin^2(\pi\|x-x'\|/p)}{\ell^2}\right)$	Captures periodicity
Linear	$\sigma_b^2 + \sigma_v^2(\mathbf{x} \cdot \mathbf{x}')$	Bayesian linear regression

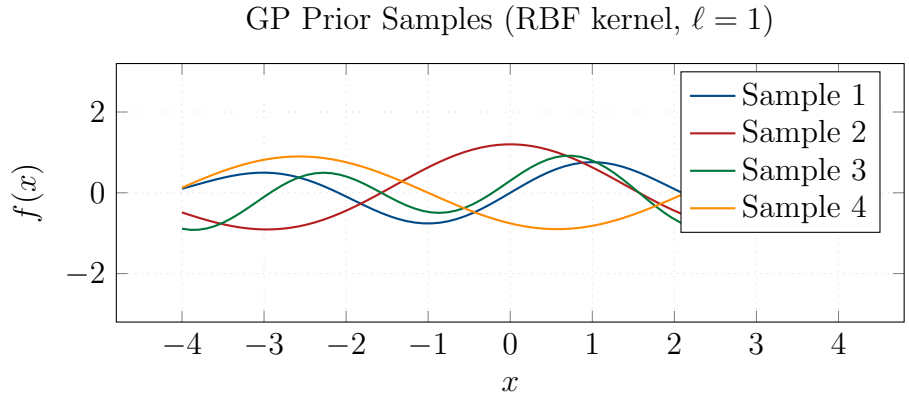


Figure 2.1: Samples from a GP prior with RBF kernel and shaded $\pm 2\sigma$ uncertainty band.

2.3 Randomized Methods in Machine Learning

2.3.1 Random Projections (Johnson–Lindenstrauss)

Theorem 2.3.1: Johnson–Lindenstrauss Lemma

For any $0 < \epsilon < 1$ and n points in $\mathbb{R}^d$, a random projection $\mathbf{A} \in \mathbb{R}^{k \times d}$ with $k = O(\log n / \epsilon^2)$ satisfies $(1-\epsilon)\|\mathbf{u}-\mathbf{v}\|^2 \leq \|\mathbf{A}\mathbf{u} - \mathbf{A}\mathbf{v}\|^2 \leq (1+\epsilon)\|\mathbf{u}-\mathbf{v}\|^2$ with high probability.

2.3.2 Random Features for Kernel Approximation

Bochner's theorem: any shift-invariant kernel can be written $k(\mathbf{x}, \mathbf{x}') = \mathbb{E}_\omega[e^{j\omega^\top(\mathbf{x}-\mathbf{x}')}]$. Approximate with D random Fourier features:

$$k(\mathbf{x}, \mathbf{x}') \approx \phi(\mathbf{x})^\top \phi(\mathbf{x}'), \quad \phi(\mathbf{x}) = \tfrac{1}{\sqrt{D}}[\cos(\omega_i^\top \mathbf{x} + b_i)]_{i=1}^D,$$

reducing kernel machine complexity from $O(N^2)$ to $O(ND)$.

2.3.3 Stochastic Gradient Descent (SGD)

Mini-batch SGD approximates the gradient using a batch $\mathcal{B} \subset [N]$:

$$\theta \leftarrow \theta - \eta \frac{N}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla_\theta \ell_i(\theta).$$

Variants include Adam, AdaGrad, RMSProp, each adapting per-parameter learning rates.

Chapter 3

Bayesian Neural Networks and Approximate Inference

3.1 Bayesian Neural Networks (BNNs)

Definition 3.1.1: Bayesian Neural Network

A BNN places a prior distribution $p(\mathbf{W})$ over network weights instead of treating them as point estimates. The posterior after observing data is:

$$p(\mathbf{W} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \mathbf{W})\, p(\mathbf{W})}{p(\mathcal{D})}.$$

Predictions are made by *Bayesian model averaging*:

$$p(y^* \mid \mathbf{x}^*, \mathcal{D}) = \int p(y^* \mid \mathbf{x}^*, \mathbf{W})\, p(\mathbf{W} \mid \mathcal{D})\, d\mathbf{W}.$$

3.1.1 Benefits and Challenges

- **Benefits:** Principled uncertainty quantification; resistance to overfitting; natural regularisation via priors.
- **Challenges:** The posterior $p(\mathbf{W} \mid \mathcal{D})$ is intractable for modern networks; requires approximate inference.

3.2 Approximate Inference

3.2.1 Markov Chain Monte Carlo (MCMC)

MCMC constructs a Markov chain whose stationary distribution is the target posterior. **Hamiltonian Monte Carlo (HMC)** uses gradient information to propose distant samples with high acceptance probability:

$$H(\mathbf{q}, \mathbf{p}) = U(\mathbf{q}) + K(\mathbf{p}), \quad U(\mathbf{q}) = -\log p(\mathbf{q} \mid \mathcal{D}), \quad K(\mathbf{p}) = \tfrac{1}{2}\mathbf{p}^\top M^{-1}\mathbf{p}.$$

3.2.2 Variational Inference (VI)

VI approximates the intractable posterior $p(\theta \mid \mathcal{D})$ by optimising a simpler distribution $q_\phi(\theta)$ to minimise the KL divergence:

$$q_\phi^* = \arg\min_{q \in \mathcal{Q}} \mathrm{KL}[q_\phi(\theta) \parallel p(\theta \mid \mathcal{D})] \,.$$

This is equivalent to maximising the **Evidence Lower BOund (ELBO)**:

$$\mathcal{L}(\phi) = \underbrace{\mathbb{E}_{q_\phi}[\log p(\mathcal{D} \mid \theta)]}_{\text{expected log-likelihood}} - \underbrace{\mathrm{KL}[q_\phi(\theta) \| p(\theta)]}_{\text{regularisation}} .$$

3.2.3 Monte Carlo Dropout

Dropout at test time (Gal & Ghahramani, 2016) provides a cheap BNN approximation: run T stochastic forward passes and use sample mean/variance as predictive mean/uncertainty:

$$\hat{y} \approx \frac{1}{T} \sum_{t=1}^{T} f_{\hat{\mathbf{W}}_t}(\mathbf{x}^*), \qquad \hat{\mathbf{W}}_t \sim q_\phi(\mathbf{W}).$$

Chapter 4

Variational Autoencoders and Generative Models

4.1 Variational Autoencoders (VAEs)

Definition 4.1.1: VAE

A VAE is a latent-variable generative model $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x} \mid \mathbf{z})\, p(\mathbf{z})$ where $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and the likelihood $p_\theta(\mathbf{x} \mid \mathbf{z})$ is a neural decoder. Inference uses an amortised encoder $q_\phi(\mathbf{z} \mid \mathbf{x})$.

4.1.1 The ELBO Objective

The VAE maximises:

$$\mathcal{L}(\theta, \phi; \mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x} \mid \mathbf{z})]}_{\text{reconstruction}} - \underbrace{\mathrm{KL}[q_\phi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})]}_{\text{regularisation}}. \tag{4.1}$$

4.1.2 Reparameterisation Trick

The key trick enabling backpropagation through stochastic nodes:

$$\mathbf{z} = \mu_\phi(\mathbf{x}) + \boldsymbol{\sigma}_\phi(\mathbf{x}) \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

4.2 Generative Models

4.2.1 Generative Adversarial Networks (GANs)

A GAN comprises a generator G and discriminator D playing a minimax game:

$$\min_G \max_D \; \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))].$$

4.2.2 Normalising Flows

Learn an invertible mapping $f : \mathcal{Z} \to \mathcal{X}$; exact log-likelihood via the change-of-variables formula:

$$\log p(\mathbf{x}) = \log p_{\mathbf{z}}(f^{-1}(\mathbf{x})) + \log \left| \det \frac{\partial f^{-1}}{\partial \mathbf{x}} \right|.$$

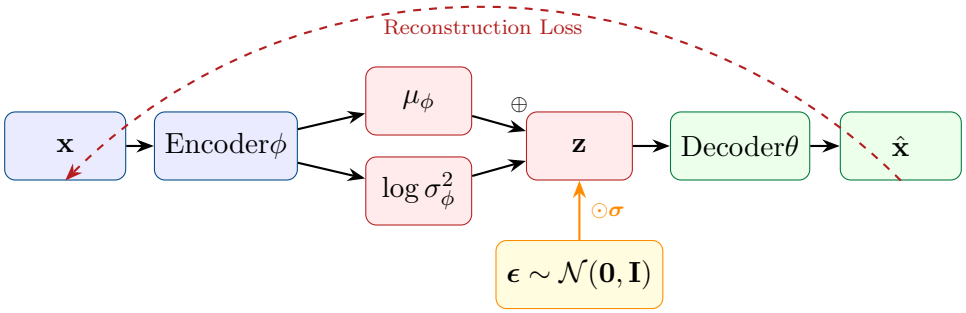


Figure 4.1: VAE architecture showing the encoder, reparameterisation trick, decoder, and training signal flow.

4.2.3 Diffusion Models

Forward process adds Gaussian noise over T steps: $q(\mathbf{x}_t \mid \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\,\mathbf{x}_{t-1}, \beta_t \mathbf{I})$. A neural network learns the reverse denoising process $p_\theta(\mathbf{x}_{t-1} \mid \mathbf{x}_t)$.

Model	**Exact LL**	**Sample Quality**	**Training Stability**
VAE	lower bound	moderate	stable
GAN	no	high	unstable (mode collapse)
Flow	yes	moderate	stable
Diffusion	lower bound	very high	stable

Part III

NLP and Deep Sequence Models

Chapter 5

Recurrent Neural Networks

5.1 Vanilla RNN

Definition 5.1.1: Recurrent Neural Network

An RNN maintains a hidden state $\mathbf{h}_t$ updated at each time step:

$$\mathbf{h}_t = \tanh(\mathbf{W}_h \mathbf{h}_{t-1} + \mathbf{W}_x \mathbf{x}_t + \mathbf{b}_h), \qquad \mathbf{y}_t = \mathbf{W}_y \mathbf{h}_t + \mathbf{b}_y.$$

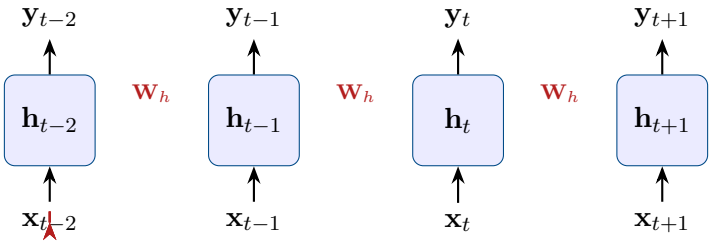


Figure 5.1: Unrolled RNN across four time steps; $\mathbf{W}_h$ is shared.

5.2 Backpropagation Through Time (BPTT)

Gradients are computed by unrolling the network and applying the chain rule. The gradient of loss $\mathcal{L}$ w.r.t. $\mathbf{W}_h$ is:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_h} = \sum_{t=1}^{T} \frac{\partial \mathcal{L}_t}{\partial \mathbf{h}_t} \sum_{k=1}^{t} \left(\prod_{j=k+1}^{t} \frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}} \right) \frac{\partial \mathbf{h}_k}{\partial \mathbf{W}_h}.$$

5.2.1 Vanishing and Exploding Gradients

The Jacobian product $\prod_j \frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}}$ causes gradients to either vanish ($\to 0$) or explode ($\to \infty$) exponentially in the sequence length. Gradient clipping addresses explosion: $\hat{g} \leftarrow \min\left(1, \frac{\tau}{\|g\|}\right) g$.

5.3 Long Short-Term Memory (LSTM)

The LSTM (Hochreiter & Schmidhuber, 1997) introduces a *cell state* $\mathbf{c}_t$ with gating mechanisms to carry information over long sequences.

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_f) \quad \text{(forget gate)} \tag{5.1}$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_i) \quad \text{(input gate)} \tag{5.2}$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_c) \quad \text{(candidate cell)} \tag{5.3}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad \text{(cell update)} \tag{5.4}$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}; \mathbf{x}_t] + \mathbf{b}_o) \quad \text{(output gate)} \tag{5.5}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad \text{(hidden state)} \tag{5.6}$$

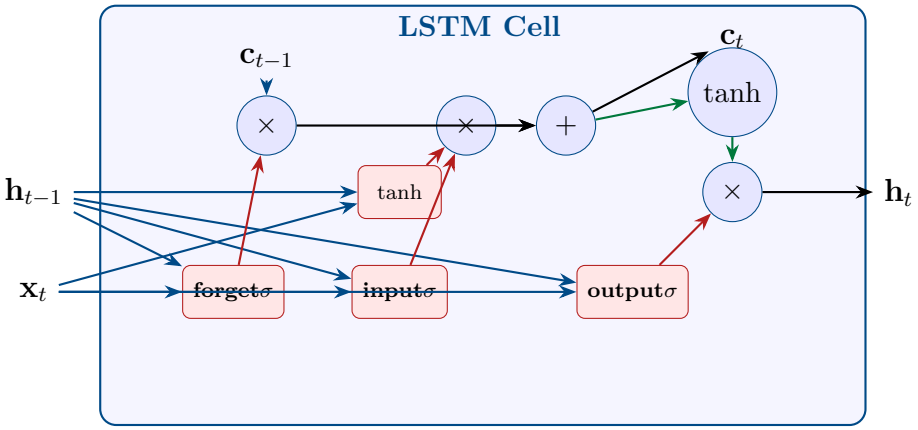


Figure 5.2: LSTM cell with forget, input, and output gates, plus cell state highway.

Chapter 6

Attention, Memory, and Neural Turing Machines

6.1 Attention Mechanisms

Definition 6.1.1: Attention

Given queries $\mathbf{Q}$, keys $\mathbf{K}$, values $\mathbf{V}$, *Scaled Dot-Product Attention* computes:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}.$$

The scaling factor $1/\sqrt{d_k}$ stabilises gradients.

6.1.1 Multi-Head Attention (Transformer)

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\,\mathbf{W}^O,$$

where $\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V)$.

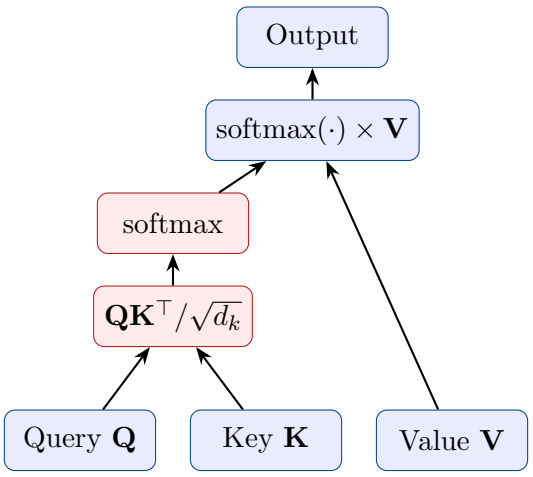


Figure 6.1: Scaled dot-product attention computation graph.

6.2 Memory Networks

Memory Networks (Weston et al., 2015) augment neural networks with an external memory matrix $\mathbf{M} \in \mathbb{R}^{N_m \times d}$. At inference, the model computes soft attention over memory slots:

$$p_i = \text{softmax}(\mathbf{q}^\top \mathbf{m}_i), \qquad \mathbf{o} = \sum_i p_i \, \mathbf{c}_i,$$

where $\mathbf{q}$ is a query, $\mathbf{m}_i$ are input embeddings and $\mathbf{c}_i$ are output embeddings of memory slots. **End-to-End Memory Networks** stack multiple memory hops for iterative reasoning: $\mathbf{u}^{k+1} = \mathbf{u}^k + \mathbf{o}^k$.

6.3 Neural Turing Machines (NTMs)

An NTM (Graves et al., 2014) couples an LSTM controller with an external memory bank $\mathbf{M}_t \in \mathbb{R}^{N \times W}$. Read/write heads produce attention weights $\mathbf{w}_t \in \Delta^{N-1}$:

$$\mathbf{r}_t = \sum_i w_t(i) \, \mathbf{M}_t(i), \quad \text{(read)}$$

$$\mathbf{M}_t(i) = \mathbf{M}_{t-1}(i) \, [1 - w_t^w(i) \, e_t] + w_t^w(i) \, \mathbf{a}_t. \quad \text{(write)}$$

Addressing combines:

1. *Content-based*: cosine similarity between key $\mathbf{k}_t$ and rows of $\mathbf{M}_t$.
2. *Location-based*: convolutional shift $\mathbf{w}_t^g = \sum_j s_t(j) \, \tilde{w}_{(i-j)}$.

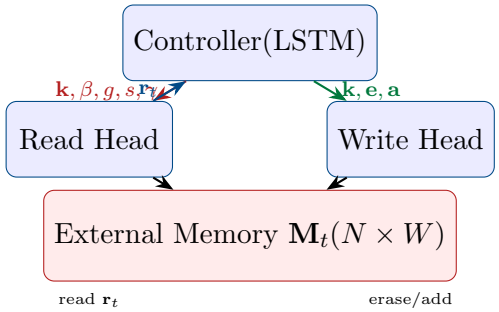


Figure 6.2: Neural Turing Machine: controller interacts with external memory via differentiable read/write heads.

Chapter 7

NLP Applications

7.1 Machine Translation

7.1.1 Encoder–Decoder Architecture

$$\mathbf{h}_t^{\text{enc}} = \text{LSTM}_{\text{enc}}(\mathbf{x}_t, \mathbf{h}_{t-1}^{\text{enc}}), \quad \mathbf{s}_j = \text{LSTM}_{\text{dec}}(\mathbf{s}_{j-1}, \mathbf{c}_j),$$

$$\mathbf{c}_j = \sum_t \alpha_{jt}\, \mathbf{h}_t^{\text{enc}}, \quad \alpha_{jt} = \frac{\exp(e_{jt})}{\sum_{t'} \exp(e_{jt'})}, \quad e_{jt} = \mathbf{v}_a^\top \tanh(\mathbf{W}_a \mathbf{s}_{j-1} + \mathbf{U}_a \mathbf{h}_t).$$

7.1.2 Transformer for Translation

The Transformer (Vaswani et al., 2017) replaces recurrence entirely:

- **Encoder:** N_e layers of Multi-Head Self-Attention + FFN.
- **Decoder:** N_d layers of Masked MH-Self-Attn + Cross-Attn + FFN.
- **Positional Encoding:** $\text{PE}(pos, 2i) = \sin(pos/10000^{2i/d_{\text{model}}})$.

7.2 Question Answering (QA)

- **Extractive QA** (e.g. SQuAD): predict start/end span indices in passage.
- **Abstractive QA**: generate a free-form answer (seq2seq).
- **BERT-based QA**: fine-tune pretrained transformer; output heads predict $P(\text{start})$ and $P(\text{end})$.
- **Retrieval-Augmented Generation (RAG)**: retrieve relevant documents then generate answer conditioned on retrieved context.

7.3 Speech Recognition

Automatic Speech Recognition (ASR) maps an acoustic feature sequence $\mathbf{o}_{1:T}$ (e.g. log-mel filterbanks) to a word sequence $w_{1:L}$.

7.3.1 Connectionist Temporal Classification (CTC)

CTC marginalises over all valid alignments π of length T:

$$p(\mathbf{y} \mid \mathbf{x}) = \sum_{\pi:\mathcal{B}(\pi)=\mathbf{y}} \prod_{t=1}^{T} p(\pi_t \mid \mathbf{x}),$$

where $\mathcal{B}$ collapses repeated tokens and removes blanks. Forward–backward dynamic programming computes this efficiently.

7.3.2 Attention-Based Encoder–Decoder (AED)

$$s_l, c_l = \text{AttDecoder}(s_{l-1}, y_{l-1}, \mathbf{h}_{1:T}), \quad y_l \sim P_\theta(\cdot \mid s_l, c_l).$$

7.3.3 Whisper / wav2vec 2.0

Modern end-to-end models use self-supervised pretraining on large unlabelled audio corpora, followed by supervised fine-tuning, achieving near-human performance on standard benchmarks.

7.4 Syntactic and Semantic Parsing

7.4.1 Syntactic Parsing

Constituency parsing produces a phrase-structure tree; **dependency parsing** labels directed arcs between words. Neural approaches use:

- **BiLSTM + CRF**: contextualised token representations + structured output.
- **Graph Neural Networks (GNNs)**: encode tree structure explicitly.
- **Transformer span-based**: score all spans via self-attention representations.

7.4.2 Semantic Parsing

Maps natural language to a formal meaning representation (e.g. SQL, SPARQL, λ-calculus, AMR). Sequence-to-sequence models with copy mechanisms (pointer networks) handle novel entities. **T5/CodeT5** achieve state-of-the-art on Text-to-SQL tasks.

Chapter 8

GPU Optimisation for Neural Networks

8.1 Hardware Landscape

Modern GPUs (e.g. NVIDIA A100/H100) provide:

- Thousands of CUDA cores for parallel floating-point ops.
- Tensor Cores: hardware-accelerated mixed-precision matrix multiplication.
- High-bandwidth memory (HBM): >2 TB/s bandwidth (H100 SXM).

8.2 Memory Optimisation

- **Mixed Precision (FP16/BF16):** halves memory; use FP32 master weights.
- **Gradient Checkpointing:** recompute activations during backward pass to reduce peak memory by $O(\sqrt{n})$ at cost of $O(1)$ extra forward pass.
- **Activation Offloading:** swap tensors to CPU/NVMe when idle.
- **FlashAttention:** fuses attention operations to avoid $O(L^2)$ memory materialisation; operates in SRAM tiles for $O(L)$ memory.

8.3 Compute Optimisation

- **Data parallelism:** replicate model; aggregate gradients via AllReduce.
- **Model parallelism:** partition layers across GPUs (tensor/pipeline parallel).
- **Fused kernels:** combine elementwise ops (e.g. layer-norm + GELU) to reduce kernel launch overhead.
- **Quantisation:** INT8/INT4 weights reduce memory and increase throughput (e.g. GPTQ, AWQ).
- **Compilation (torch.compile / XLA):** static graph optimisation, operator fusion, and autotuned tile sizes.

8.4 Distributed Training at Scale

The NCCL (NVIDIA Collective Communications Library) implements efficient multi-GPU communication primitives. **ZeRO** (Zero Redundancy Optimizer) partitions optimizer states, gradients, and parameters across GPUs to scale to trillion-parameter models. The total communication volume per step is $O(3\Psi)$ for ZeRO-3 vs. $O(2\Psi)$ for standard data parallelism, where Ψ is model size.

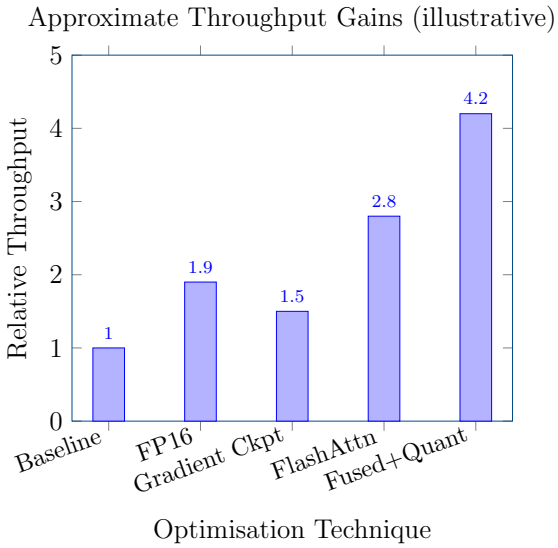


Figure 8.1: Illustrative relative throughput gains from common GPU optimisation techniques applied cumulatively.

Key References

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